

SALITHA MARASINGHE

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SUMMARY

Final-year undergraduate with hands-on industry experience as a Data Engineer intern, bringing a track record of delivering end-to-end data solutions across data engineering and cloud platforms, and applies end-to-end ownership mindset and cross-functional Agile collaboration to deliver data solutions across the full project lifecycle.

SKILLS

Languages: Python | SQL | PySpark

Cloud & Platforms: Microsoft Azure | Microsoft Fabric | Databricks

Data Engineering & Orchestration Languages: Azure Data Factory | Fabric Data Factory | Microsoft SSIS | ETL/ELT Pipeline Design | Medallion Architecture | Apache Spark

Data Warehousing & BI: SQL Server | SSAS | Power BI | Star & Snowflake Schema Design

Databases & Storage: SQL Server | MySQL | Firebase | MongoDB

Software & DevOps: Fast API | React | GitHub | BitBucket | Railway | Firebase

ML & AI: NumPy | Pandas | Scikit-Learn | Matplotlib | Time Series Forecasting | AI integration

Methodologies & Practices: Technical Team Leadership | End-to-End System Ownership | Cross-Functional Agile Collaboration | Proactive Problem Solving

EXPERIENCE

Data Engineer Intern • TRACE – Sri Lanka | Remote • Aug 2025 – Feb 2026

- Built an automated Python ETL pipeline (Customs Data Transformer) to ingest, validate, and transform Excel manifests into standards-compliant XML, eliminating a manual enterprise data submission process.
- Designed ER schemas and implemented the backend for a multi-source B2B pricing platform (Wasi Lanka) and conducted market research on feasibility to support strategic decision making.
- Architected an LLM-powered AI data agent (Tourism AI Agent) with MCP-agent design and consent-based behavioral data collection pipelines.
- Adhered to DevOps principles by enforcing Git/BitBucket version control, rigorous code review standards, and continuous integration workflows across codebases.
- Acted as team lead within a cross-functional Agile delivery environment, managing full-lifecycle sprint deliverables from development and QA through to deployment, while translating complex data pipeline architectures into clear outputs for non-technical business stakeholders.

PROJECTS

ISP Customer Analytics & Churn Prediction Platform

Tech Stack: Azure Data Factory, Databricks, PySpark, SQL, Scikit-learn, Power BI

- Engineered an end-to-end orchestrated cloud Lakehouse pipeline ingesting multi-source data into distributed object storage (ADLS Gen2), implementing complex distributed data transformations using Apache Spark (via Databricks) through a Bronze/Silver/Gold Medallion architecture.
- Designed a Star schema dimensional model in the Gold layer and exposed curated Delta tables via Databricks SQL Warehouse with live Power BI connectivity, enabling real-time dashboard updates across customer health, network SLA performance, and support and billing risk reports.
- Engineered a customer-level analytical feature dataset from complex network, support, and billing signals, integrating predictive outputs back into the Lakehouse architecture to support downstream churn risk dashboarding in Power BI.
- Orchestrated the full pipeline via Azure Data Factory, automating data extraction through Databricks ETL and ML execution to live Power BI dashboards.

Earthquake Event Analysis

Tech Stack: Microsoft Fabric, Microsoft Data Factory, PySpark, Power BI

- Built an automated medallion architecture pipeline on Microsoft Fabric ingesting raw USGS earthquake data via API into a Data Lake, transforming JSON through Bronze/Silver/Gold layers using PySpark, and enriching the Gold layer with reverse geocoding for location resolution.
- Orchestrated daily pipeline refresh via Microsoft Data Factory and delivered interactive Power BI dashboards covering global earthquake distribution, magnitude comparisons, and significance analysis by country.

Hospital Data Warehouse Management

Tech Stack: Microsoft SSIS, Microsoft SSAS, SQL Server, SQL, Power BI

- Designed a hospital management data warehouse using a snowflake schema with SCD Type 2 for historical location tracking.
- Built an SSIS ETL pipeline extracting from multi-source CSV and database inputs, applying transformations via stored procedures through a staging layer into the warehouse.
- Created an SSAS multidimensional OLAP cube and developed Power BI reports with drill-through, trend analysis, and slicers for patient admissions and doctor performance analytics.

CropIQ Smart Soil Health Prediction System

Tech Stack: Python, Firebase Realtime Database, Scikit-learn, Fast API, React

- Engineered a real-time IoT data ingestion pipeline (Hardware + Integration), securely processing continuous environmental sensor telemetry (NPK, temperature, humidity, soil moisture) into Firebase Realtime Database.
- Developed a full-stack application (Fast API Backend with React Frontend) to expose real-time sensor history, aggregated statistics, Waterlogging Risk Prediction and a Gemini Integrated AI assistant chatbot.

Sri Lanka Tourism Intelligence Platform | Research Project

Tech Stack: Python, Fast API, React, Firebase, Railway

- Engineered a multi-source data ingestion pipeline integrating macroeconomic indicators, behavioral signals, and weather covariates to support predictive analytics models (arrival forecasting, Revenue Forecasting, source market segmentation) and developed a scalable Fast API backend to serve modeled analytical outputs to downstream client applications and dashboards.
- Engineered three Groq LLM geopolitical agents with automated weekly refresh, independently analyzing global event impact on arrival forecasts, revenue risk, and source market intelligence to surface actionable recommendations for strategic decision-making.

EDUCATION

BSc (Hons) Information Technology – Specializing in Data Science (Undergraduate)

Sri Lanka Institute of Information Technology (SLIIT) | 2022 – Present

- Relevant Coursework: Database Warehousing and Business Intelligence (DWBI), Machine Learning and Optimization Methods (MLOM), Theory and Practices in Statistical Modelling (TPSM), Introduction to Internet of Things and Big Data Analytics (IoTBD)

REFERENCES

Lahiru Wijesekara

Senior Solutions Architect | Expert in Enterprise Applications & Integration | Software Consultant | Academic & Industry Researcher

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